SG No. 122 D_{2d}^{12} $I\bar{4}2d$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `4a`, site symmetry: `-4..`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `-4..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[1,2,-3,-4]`
2	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,6,-7,-8]`

Table 2: Wyckoff bond: `8b@4a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ 0]$	`[1,-2]`
2	$[Y,\ -X,\ 0]$	$[0,\ 0,\ 0]$	`[3,-4]`
3	$[-X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,-6]`
4	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7,-8]`

Table 3: Wyckoff bond: `16c@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$[0,\ 0,\ 0]$	`[2]`
3	$[Y,\ -X,\ -Z]$	$[0,\ 0,\ 0]$	`[3]`
4	$[-Y,\ X,\ -Z]$	$[0,\ 0,\ 0]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7]`
8	$[Y,\ X,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[8]`

* Wyckoff site: `4b`, site symmetry: `-4..`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `-4..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,-3,-4]`
2	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[5,6,-7,-8]`

Table 5: Wyckoff bond: `8b@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-2]`
2	$[Y,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,-4]`
3	$[-X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[5,-6]`
4	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[7,-8]`

Table 6: Wyckoff bond: `16c@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2]`
3	$[Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3]`
4	$[-Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[7]`
8	$[Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[8]`

* Wyckoff site: `8c`, site symmetry: `2..`

Table 7: Wyckoff bond: `8a@8c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[Y, -X, 0]$	$[0, 0, -z]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4} - z\right]$	`[5,-6]`
4	$[-Y, -X, 0]$	$\left[\frac{1}{2}, 0, z + \frac{3}{4}\right]$	`[7,-8]`

Table 8: Wyckoff bond: `8b@8c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2]`
2	$[0, 0, -Z]$	$[0, 0, -z]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4} - z\right]$	`[5,6]`
4	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z + \frac{3}{4}\right]$	`[7,8]`

Table 9: Wyckoff bond: `16c@8c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[Y, -X, -Z]$	$[0, 0, -z]$	`[3]`
4	$[-Y, X, -Z]$	$[0, 0, -z]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4} - z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4} - z\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{1}{2}, 0, z + \frac{3}{4}\right]$	`[7]`
8	$[Y, X, Z]$	$\left[\frac{1}{2}, 0, z + \frac{3}{4}\right]$	`[8]`

* Wyckoff site: `8d`, site symmetry: `.2.`

Table 10: Wyckoff bond: `8a@8d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0,\ X,\ Z]$	$\left[x,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[1,-6]`
2	$[0,\ -X,\ Z]$	$\left[-x,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[2,-5]`
3	$[X,\ 0,\ -Z]$	$\left[\frac{1}{4},\ -x,\ \frac{7}{8}\right]$	`[3,-7]`
4	$[-X,\ 0,\ -Z]$	$\left[\frac{3}{4},\ x,\ \frac{7}{8}\right]$	`[4,-8]`

Table 11: Wyckoff bond: `8b@8d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[x,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[1,6]`
2	$[-X,\ 0,\ 0]$	$\left[-x,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[2,5]`
3	$[0,\ -X,\ 0]$	$\left[\frac{1}{4},\ -x,\ \frac{7}{8}\right]$	`[3,7]`
4	$[0,\ X,\ 0]$	$\left[\frac{3}{4},\ x,\ \frac{7}{8}\right]$	`[4,8]`

Table 12: Wyckoff bond: `16c@8d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[-x,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[2]`
3	$[Y,\ -X,\ -Z]$	$\left[\frac{1}{4},\ -x,\ \frac{7}{8}\right]$	`[3]`
4	$[-Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ x,\ \frac{7}{8}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[-x,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ -x,\ \frac{7}{8}\right]$	`[7]`
8	$[Y,\ X,\ Z]$	$\left[\frac{3}{4},\ x,\ \frac{7}{8}\right]$	`[8]`

* Wyckoff site: `16e`, site symmetry: `1`

Table 13: Wyckoff bond: `16a@16e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$[-x,\ -y,\ z]$	`[2]`
3	$[Y,\ -X,\ -Z]$	$[y,\ -x,\ -z]$	`[3]`
4	$[-Y,\ X,\ -Z]$	$[-y,\ x,\ -z]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ y,\ \frac{3}{4}-z\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[x+\frac{1}{2},\ -y,\ \frac{3}{4}-z\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{2}-y,\ -x,\ z+\frac{3}{4}\right]$	`[7]`
8	$[Y,\ X,\ Z]$	$\left[y+\frac{1}{2},\ x,\ z+\frac{3}{4}\right]$	`[8]`